

Calculus (PR12)

Week 5: Definite Integrals and Differential Equations

Announcement

!! Deadline:

Submit your research report draft by Friday 23:59.

Learning Goals



Accumulated Change

- Apply **definite integration** to quantify accumulated change.
- **Split** complex data using **linearity** and **interval additivity**.



FTC

- Understand concept behind the **fundamental theorem of calculus**.
- Compute **total system shifts** using **boundary values** and antiderivatives.



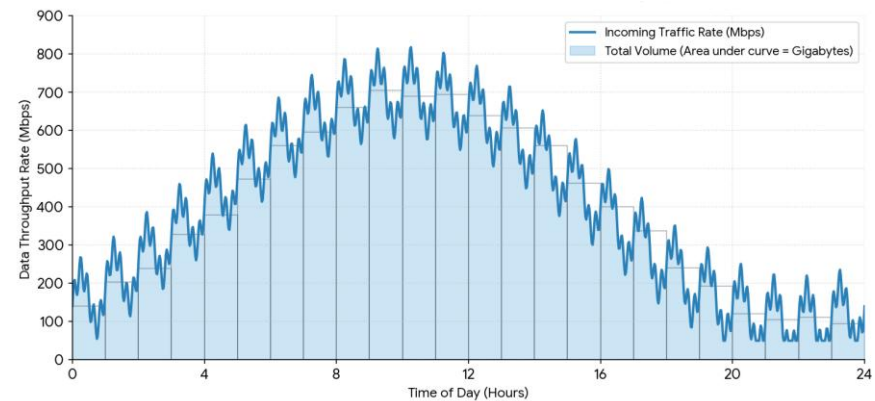
Differential Equations

- Apply **variable separation** to solve **independent growth curves**.
- Use **integrating factors** to **collapse first-order linear** expressions.



Think and Share

Scenario: Your cloud server tracks incoming network traffic requests. The system provides a fluctuating rate of incoming data in Mbps.



Question: If you only have the mathematical function of data log of this changing rate, how do you compute the absolute total gigabytes processed over time?

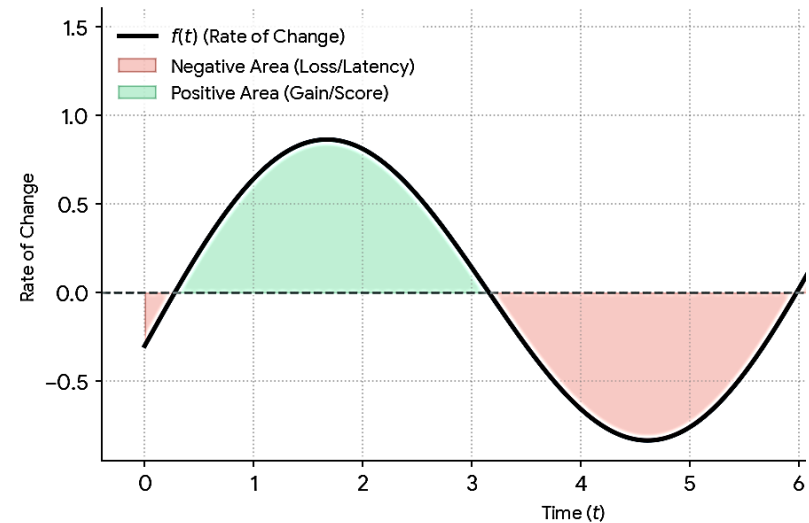


Definite Integrals as Accumulated Change

Mathematical Definition:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

x_i^* : sample point.



Intuition: The definite integral is the infinitely fine sum of tiny changes over an interval.

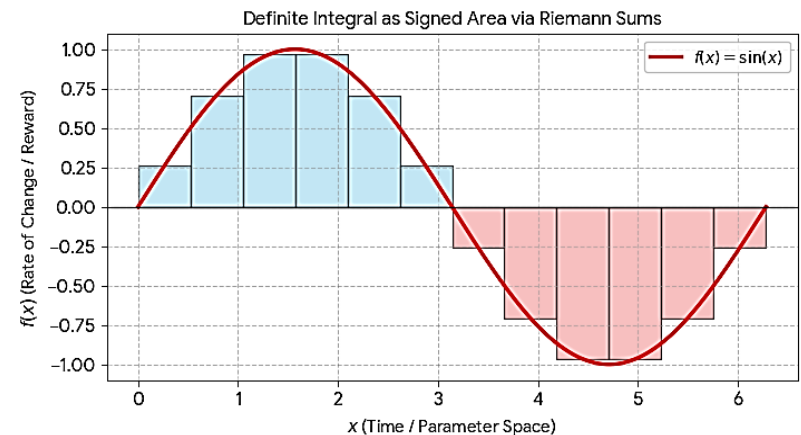


Definite Integrals as Accumulated Change

Mathematical Definition:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

x_i^* : sample point.



Intuition: The definite integral is the infinitely fine sum of tiny changes over an interval.

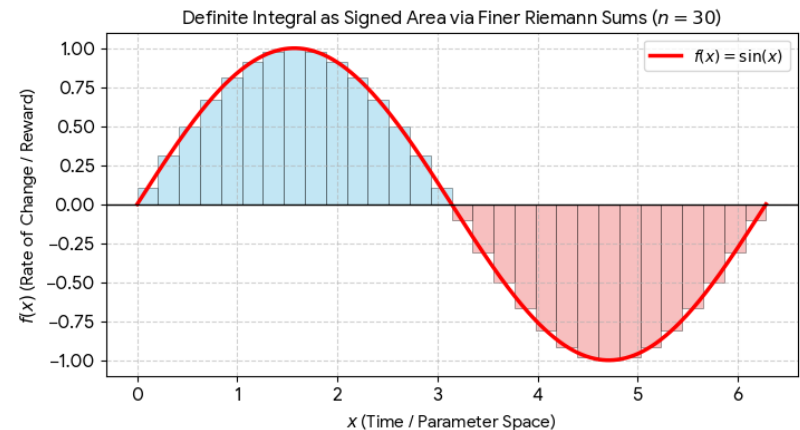


Definite Integrals as Accumulated Change

Mathematical Definition:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

x_i^* : sample point.



Intuition: The definite integral is the infinitely fine sum of tiny changes over an interval.

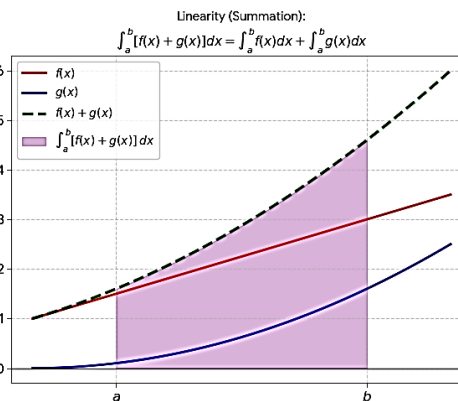
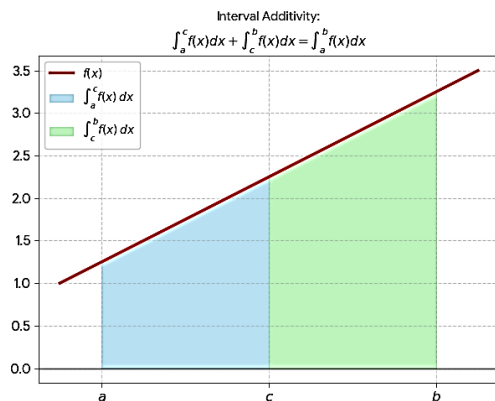
Linearity & Additivity

Linearity:

$$\int_a^b [c \cdot f(x) + d \cdot g(x)] dx = c \int_a^b f(x) dx + d \int_a^b g(x) dx$$

Interval additivity:

$$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$

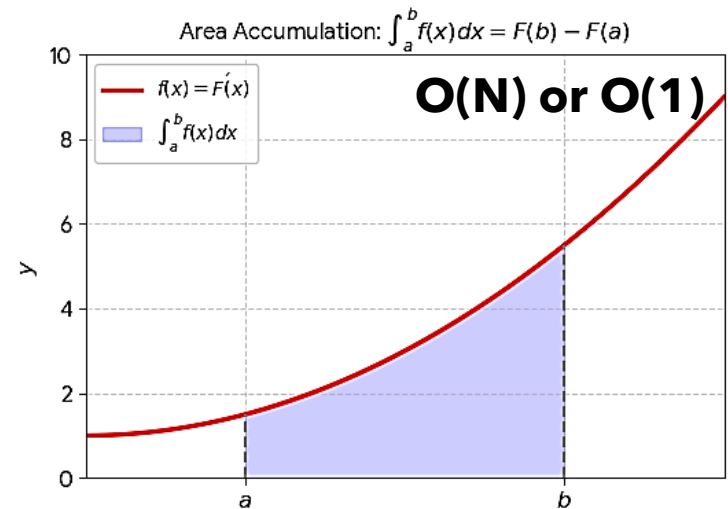


The Fundamental Theorem of Calculus

Differentiation and integration are perfect mathematical inverses.

if $F'(x) = f(x)$, then:

$$\int_a^b f(x) dx = F(b) - F(a)$$



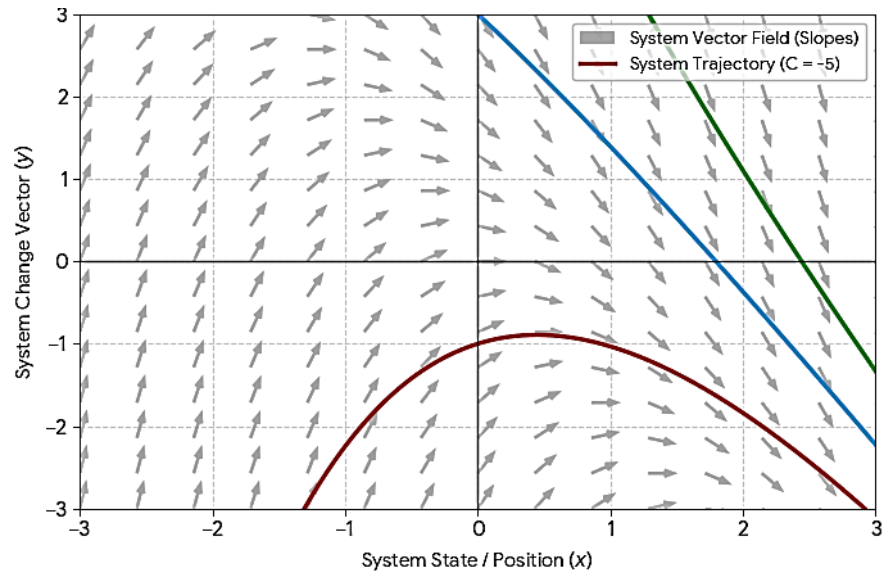
Intuition: Finding instantaneous rates and finding total sum are reversed mathematical operations.



Differential Equations (DEs)

Definition: An equation stating a structural relationship between an unknown function and its derivatives.

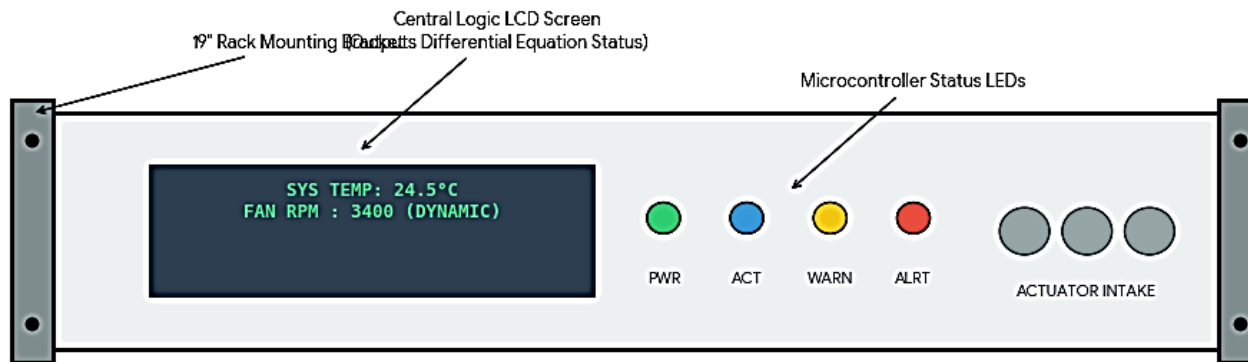
AI domain context: Gradient descent with momentum.



Example

In an automated server-room environmental controller, what dictates how fast a machine cools down?

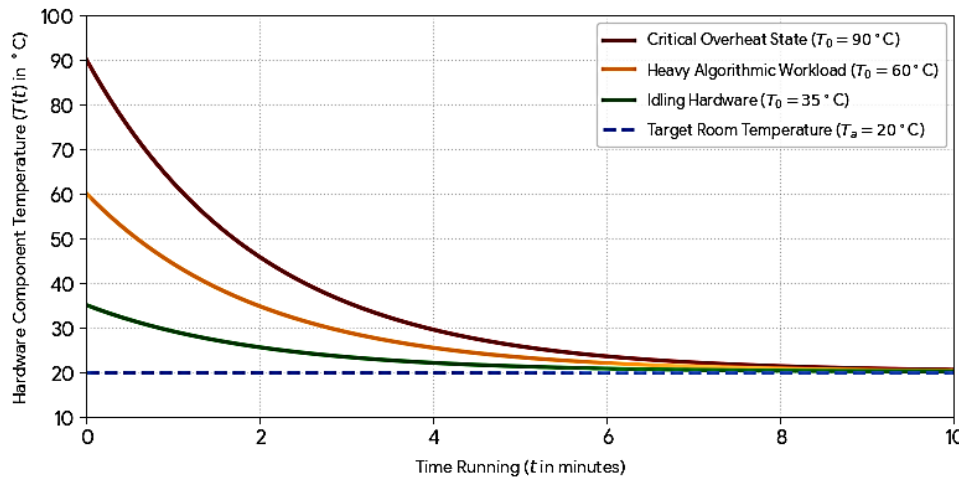
- A.** The exact temperature of the room.
- B.** The difference between the chip temperature and the ambient air temperature.



Example

In an automated server-room environmental controller, what dictates how fast a machine cools down?

- A.** The exact temperature of the room.
- B.** The difference between the chip temperature and the ambient air temperature.



First-Order Separable DEs

Standard form:

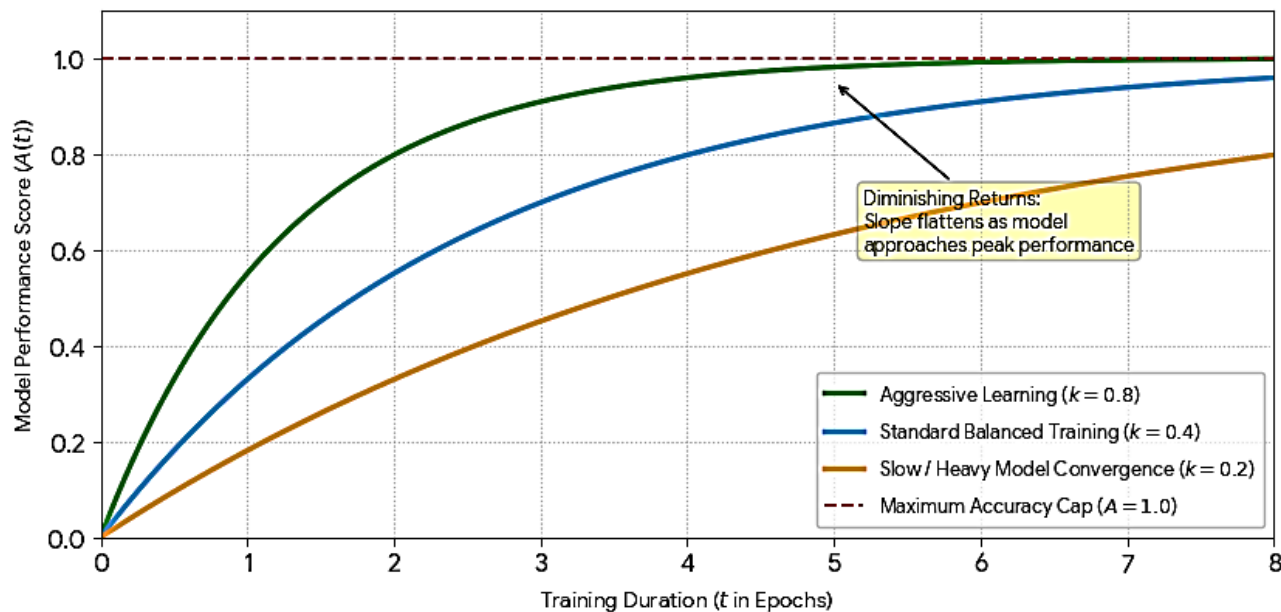
$$\frac{dy}{dx} = g(x)h(y)$$

The strategy: Group all y terms with dy on one side, and all x terms with dx on the other side. Integrate both sides independently:

$$\int \frac{1}{h(y)} dy = \int g(x) dx$$

Example - Separable DE

The scenario: An AI model is training, and its accuracy $A(t)$ grows at a rate proportional to remaining accuracy left to achieve (up to perfect score of 1.0).



Example - Separable DE

The equation:

$$\frac{dA}{dt} = k(1 - A)$$

The solution steps:

1. Separate: $\frac{1}{1-A} dA = k dt$
2. Integrate both sides: $-\ln|1 - A| = kt + C$
3. Solve for $A(t)$: $A(t) = 1 - e^{-C} e^{-kt}$



First-Order Linear DEs

Standard form:

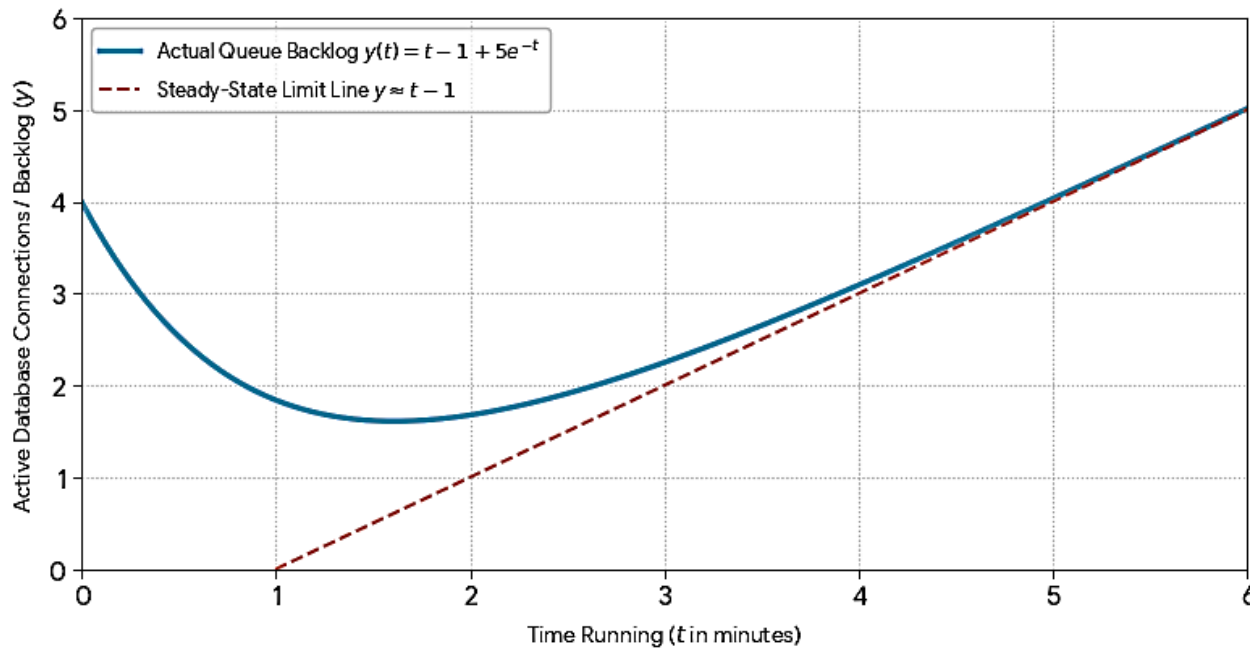
$$\frac{dy}{dx} + P(x)y = Q(x)$$

The secret weapon: The integrating factor $I(x) = e^{\int P(x)dx}$.
Multiplying the entire expression by $I(x)$ collapses the left side into a simple Product Rule derivative:

$$\frac{d}{dx} [I(x)y] = I(x)Q(x)$$

Example - Linear DE

The scenario: A cloud database queues processes requests, but the incoming traffic rate fluctuates linearly over time as $Q(t) = t$.



Example - Linear DE

The equation:

$$\frac{dy}{dt} + y = t$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$I(x) = e^{\int P(x)dx}$$

$$\frac{d}{dx} [I(x)y] = I(x)Q(x)$$

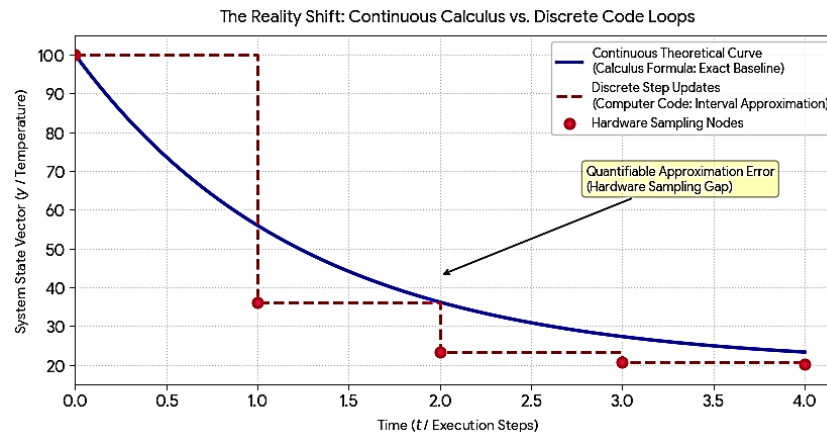
The solution steps:

1. Identify $P(t) = 1$. Compute $I(t) = e^{\int 1dt} = e^t$.
2. Multiply through: $e^t \frac{dy}{dt} + e^t y = te^t \Rightarrow \frac{d}{dt} [e^t y] = te^t$.
3. Integrate the RHS by parts $e^t y = te^t - e^t + C$.
4. Exact closed solution: $y(t) = t - 1 + Ce^{-t}$.



The Reality Shift: Continuous to Discrete

The engineering gap: Analytical calculus provides exact formulas, but modern hardware is finite and operates discretely.



The engineering solution: We translate continuous DEs into discrete numerical step updates.



Conceptual Wrap-UP

- **Definite Integrals:** Complete global accumulation from dynamic, fluctuating rates.
- **FTC:** Analytical shortcut to bypass heavy computer summation loops.
- **Differential equations:** Core mathematical framework for modeling systems in flux.
- **Numerical Methods:** Algorithm engine to transform continuous math into discrete code updates.

Next week:

Multivariable Functions and Partial Derivatives.



?

Questions

- **Follow-up reading: Calculus: A Complete Course, (sections 5.2 – 5.5 & 7.9)**
- **Joining the Discussion Groups and the Lab**

